



Chapter 7

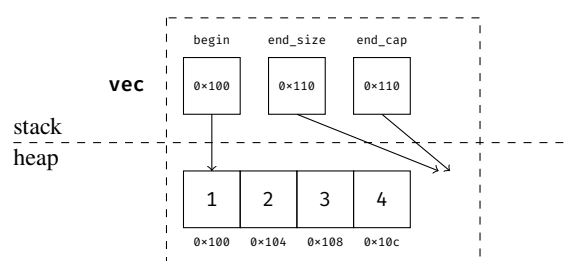
Vectors: Dynamically Resizable Arrays in C++

7.1 The STL Vector Container

If you wanted a dynamic array-based container that can grow with the size of your input, you do not have to implement it from scratch! This is thanks to the C++ **standard template library (STL)**, which provides efficient implementations of containers and algorithms for you. In this chapter, we will discuss the C++ `std::vector<>` container, which is an array-based container that manages its elements using a dynamically-sized heap-allocated C-style array. You can store elements in a vector much like how you can store elements in an array, but the vector can manage its own memory and will resize itself automatically if additional space is needed (contrary to an array, which is fixed in size). To use a vector in your program, you must `#include` the `<vector>` library. As an example, the following line declares a vector of integers:

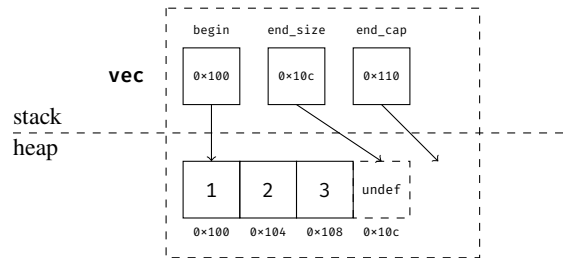
```
std::vector<int32_t> vec = {1, 2, 3, 4};
```

When the vector is initialized, it creates a C-style array on the heap (i.e., dynamic memory) with the contents {1, 2, 3, 4}. What exactly does a vector look like under the hood? This is implementation specific, but we will look at an example implementation (based off the GCC libstdc++ version of `std::vector<>`) to provide a better understanding of how they work under the hood. Here, each `std::vector<>` stores three pointers on the program stack: (1) a pointer that points to the beginning of the heap-allocated array, (2) a pointer that points one past the last *valid* element in the heap-allocated array, and (3) a pointer that points one past the last *allocated* element in the heap-allocated array. The last valid element and the last allocated element need *not* be the same.



Note that this is only one implementation of the `std::vector<>`, and the exact implementation details of a STL vector may be platform dependent as long as the public interface of the container conforms with the C++ standard. The underlying structure of a vector can differ across different standard library implementations (e.g., an implementation could keep track of one pointer and two integers for size and capacity instead of three pointers, and that would still be valid). The purpose of this chapter is to explore the general idea of how a vector works internally, and not to delve into the details of a single implementation, so you do not need to specifically memorize this vector implementation.

Regardless of how a vector is implemented, there are two important values that the vector must be able to determine at all times. The first is its **size**, which represents the number of valid elements that the underlying array *actually* holds. The second is its **capacity**, which represents the number of elements the underlying array *is able to* hold. These two values do not need to be the same. For instance, suppose we removed the last element, 4, from the back of the previously defined vector:



Since the vector's underlying array has a fixed size, it can still hold at most 4 elements. However, only 3 elements in the vector are still valid after the last element was removed. Thus, the size of the vector is 3, while its capacity is 4.

The vector is able to calculate its size by finding the distance between the pointer to the first element and the pointer one past the last valid element (i.e., `end_size - begin` using the variable names above). It is also able to calculate its capacity by finding the distance between the first element and the pointer one past the last allocated element (i.e., `end_cap - begin`).¹

The following methods can be used to initialize a vector:

```
template <typename T>
```

```
std::vector<T>();
```

Default constructor for vector that holds elements of type T; creates an empty vector with no elements; size and capacity are initially set to 0.

```
// initializes the vector v1 with zero size and zero capacity
std::vector<int32_t> v1;
```

```
template <typename T>
```

```
std::vector<T>(std::initializer_list<T> init);
```

Initializes the vector with the contents of the initializer list.

```
// initializes v2 and v3 to have contents {1, 2, 3}, with size and capacity 3
std::vector<int32_t> v2{1, 2, 3};
std::vector<int32_t> v3 = {1, 2, 3};
```

```
template <typename T>
```

```
std::vector<T>(const std::vector<T>& other);
```

Copy constructor, copies the contents of other into the constructed vector.

```
// initializes v4 with the contents of v3, or {1, 2, 3}
std::vector<int32_t> v4{v3};
```

```
template <typename T>
```

```
std::vector<T>(size_t sz);
```

Creates a vector of sz elements, where each element is value initialized; size and capacity both initially equal to sz.

```
// initializes v5 with size and capacity 5, each element initialized to 0
std::vector<int32_t> v5(5);
```

```
template <typename T>
```

```
std::vector<T>(size_t sz, T& val);
```

Creates a vector of sz elements, where each element is initialized to a value of val; size and capacity both equal to sz.

```
// initializes v6 with size and capacity 5, each element initialized to 1
std::vector<int32_t> v6(5, 1);
```

```
template <typename T, typename InputIterator>
```

```
std::vector<T>(InputIterator begin_iter, InputIterator end_iter);
```

Creates a vector with all elements in the iterator range `[begin_iter, end_iter)` — inclusive begin but exclusive end. Both `begin_iter` and `end_iter` are input iterators.

```
// initializes v7 with first three elements of v6
// .begin() returns an iterator to the first element of the vector
std::vector<int32_t> v7{v6.begin(), v6.begin() + 3};
```

¹Pointer arithmetic is performed relative to the base type of the pointer. Since integers take up 4 bytes each, incrementing `0x100` by 1 actually returns the address `0x100 + 1 × 4 = 0x104`.

7.2 Inserting and Removing Elements

※ 7.2.1 Push Back

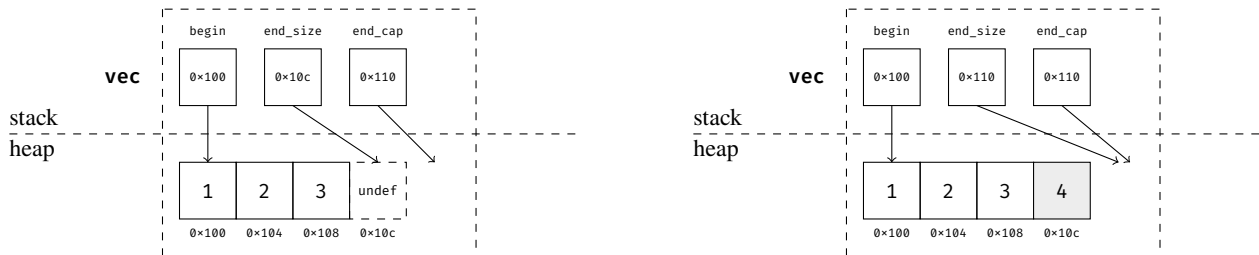
Two methods for inserting and removing elements from a vector are `std::vector::push_back()` and `std::vector::pop_back()`. When `.push_back(val)` is called, `val` is added to the back of the vector, and the size of the vector increases by one.

```
template <typename T>
```

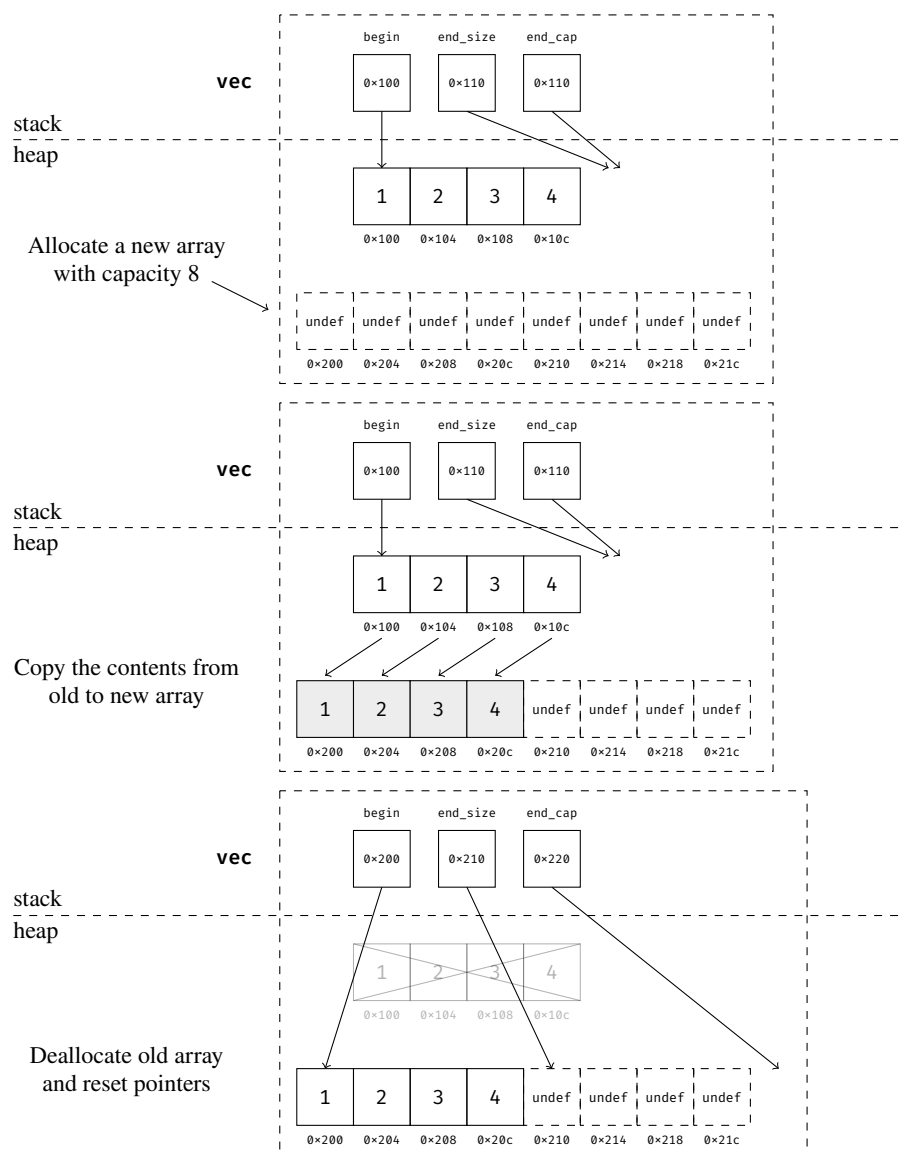
```
void std::vector<T>::push_back(const T& val);
```

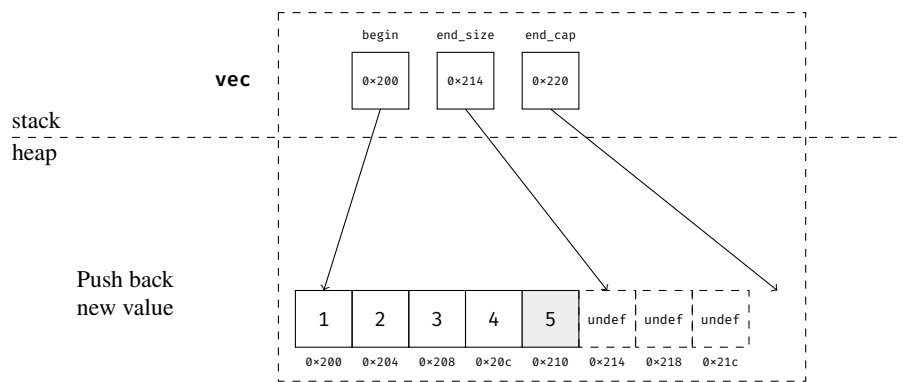
Appends a new value initialized to `val` to the back of the vector, after the current last element (if there is available capacity). If the new vector size surpasses the current capacity, the underlying array is reallocated before the new element is appended. After the object is successfully appended, the size of the vector increases by one.

Let's consider the state of the previous vector, as reproduced below on the left. If we call `vec.push_back(4)`, we push the value 4 to the back of the vector's underlying array, after the current last element (which is 3).



What happens if we try to push back another element by calling `vec.push_back(5)`? If we add this element, the size of the vector would be 5. However, the underlying heap-allocated array that stores the vector's data has a fixed size of 4, so it cannot hold a fifth element! Therefore, we will need a new array that can support a size of 5. When this happens, the vector allocates a new array in memory that is double the capacity of the original array, copies over the data, frees up the memory of the old array, and resets its internal pointers to point to the new array.



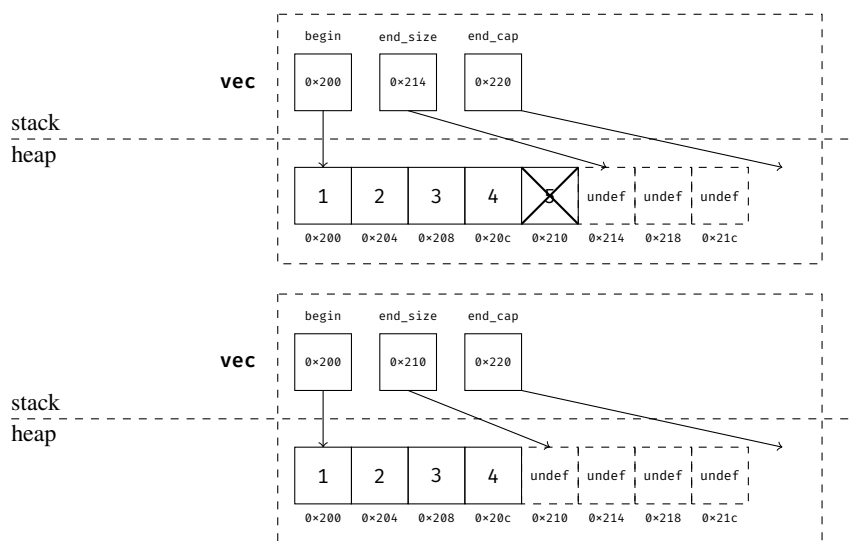


※ 7.2.2 Pop Back

To remove the last element in a vector, the `.pop_back()` method can be used.

```
template <typename T>
void std::vector<T>::pop_back();
Removes the last element in the vector, reducing its size by one.
```

If we call `vec.pop_back()` on our vector above, the 5 would be removed.



Removing an element using `.pop_back()` does not trigger a reallocation. Here, the size of the vector is 4 after the last element is removed, but the capacity is still 8. Reallocation can only happen if an insertion occurs when the vector is at capacity, or the programmer explicitly shrinks or expands the capacity of the underlying array. We will discuss several reasons why this is the case in the next section.

In addition, calling `.pop_back()` on an empty vector causes undefined behavior. The programmer is expected to know whether the vector is empty or not; this removes the need for `.pop_back()` to conduct an extra (and potentially unnecessary) "empty" check every time it is called (effectively sacrificing safety for performance).

A call to `.pop_back()` takes $\Theta(1)$ time. A call to `.push_back()` takes $\Theta(n)$ time if reallocation occurs and $\Theta(1)$ time otherwise. However, because the capacity of the underlying array is doubled upon reallocation, each $\Theta(n)$ push makes subsequent pushes less expensive; it turns out that this allows the time complexity of `.push_back()` to be treated as a $\Theta(1)$ operation if we average out all the work of reallocation across all the individual pushes. This idea is known as amortization, and we will cover it in more detail in chapter 12.

※ 7.2.3 Emplace Back

If you have a vector of large objects, `std::vector::emplace_back()` can be used as an alternative to `std::vector::push_back()`.

```
template <typename T, typename... Args>
T& std::vector<T>::emplace_back(Args&& ... args);
Appends a new element at the back of the vector, right after the current last element (if there is available capacity). If the new vector size surpasses the current capacity, the underlying array is reallocated before the new element is emplaced. The new element is constructed in place using arguments for its constructor (args), which are passed into the .emplace_back() function call. After the object is successfully emplaced, the size of the vector increases by one. Since C++17, a reference to the emplaced element is returned.
```

To illustrate how this works, consider the following `Student` object defined below:

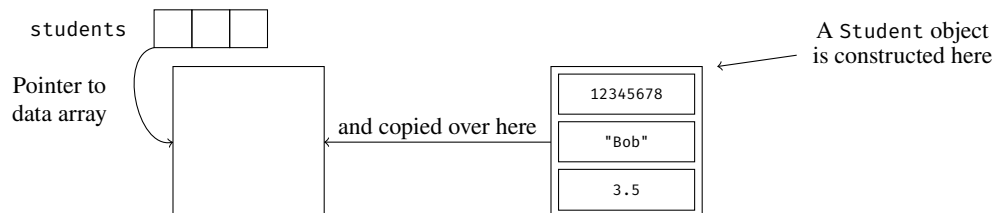
```

1 struct Student {
2     int32_t id;
3     std::string name;
4     double gpa;
5     Student(int32_t id_in, std::string name_in, double gpa_in) :
6         id{id_in}, name{name_in}, gpa{gpa_in} {}
7 };
8
9 std::vector<Student> students;
```

When you push back a `Student` object, you must first create a temporary instance of the `Student` you want to push back (as `.push_back()` can only accept an argument of the same type as the vector). Then, when `.push_back()` is invoked, the temporary `Student` is copied to the back of the vector. A visualization of this is shown below.

```

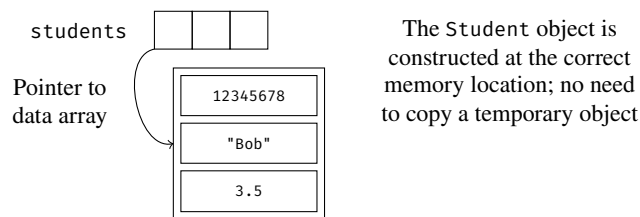
1 Student s{12345678, "Bob", 3.5};
2 students.push_back(s); // or on one line: students.push_back(Student{12345678, "Bob", 3.5});
```



What makes `.emplace_back()` different is that it accepts a list of *arguments* rather than a `Student` object itself. It then forwards the arguments to a constructor that can construct the object in the destination memory address of the vector itself. Here, `.emplace_back()` takes the three arguments, sends it to the `Student` constructor, and builds the object in the right spot. There is no need to create a temporary copy!

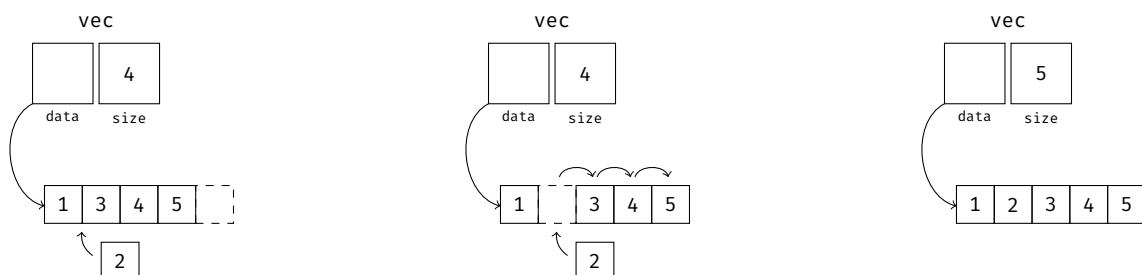
```

1 // pass into emplace_back() what you would pass into the object's constructor
2 students.emplace_back(12345678, "Bob", 3.5);
```



※ 7.2.4 Inserting and Erasing From Any Position

Appending an element to the back of a vector is easy, but inserting elements anywhere else is slightly trickier. This is because vectors require their elements to be stored contiguously in memory — if you attempt to insert an element to the vector, you must shift all the elements after the insertion point rightward to make room for the new element. The same idea applies with deletion; when you delete an element, you must shift all elements after the deletion point leftward to ensure there are no gaps in your data. Recall the illustration from the previous chapter:



Because `.push_back()`, `.emplace_back()`, and `.pop_back()` all deal with adding or removing elements at the end of the vector, there is no need to shift any elements (since there is no data after the insertion or deletion point). Thus, these operations all take $\Theta(1)$ time, assuming no reallocation is done.

If you want to insert elements elsewhere in a vector, you should use the `.insert()` function. There are several ways this can be done:

```
template <typename T>
```

```
iterator std::vector<T>::insert(const_iterator position, const T& val);
```

Inserts `val` directly before the element pointed to by the iterator `position` and returns an iterator the newly inserted element. Reallocation is done if the new size after insertion exceeds capacity.

```
1  std::vector<int32_t> vec = {0, 1, 2, 4, 5};
2  // insert 3 at index 3
3  vec.insert(vec.begin() + 3, 3);
4  // vec now {0, 1, 2, 3, 4, 5}
```

```
template <typename T>
```

```
iterator std::vector<T>::insert(const_iterator position, size_t n, const T& val);
```

Inserts `n` copies of `val` directly before the element pointed to by the iterator `position` and returns an iterator the first new element added. Reallocation is done if the new size after insertion exceeds capacity.

```
1  std::vector<int32_t> vec = {0, 1, 2, 4, 5};
2  // insert 4 copies of 3 at index 3
3  vec.insert(vec.begin() + 3, 4, 3);
4  // vec now {0, 1, 2, 3, 3, 3, 3, 4, 5}
```

```
template <typename T, typename InputIterator>
```

```
iterator std::vector<T>::insert(const_iterator position, InputIterator first, InputIterator last);
```

Inserts all elements in the iterator range `[first, last)` directly before `position` and returns an iterator the first new element added. Reallocation is done if the new size after insertion exceeds capacity.

```
1  std::vector<int32_t> vec1 = {0, 1, 5};
2  std::vector<int32_t> vec2 = {2, 3, 4};
3  // insert vec2 at index 2 of vec1
4  vec1.insert(vec1.begin() + 2, vec2.begin(), vec2.end());
5  // vec1 now {0, 1, 2, 3, 4, 5}
```

```
template <typename T>
```

```
iterator std::vector<T>::insert(const_iterator position, std::initializer_list<T> init);
```

Inserts the elements in the initializer list into the vector directly before the iterator `position` and returns an iterator to the first new element added. Reallocation is done if the new size after insertion exceeds capacity.

```
1  std::vector<int32_t> vec = {0, 1, 2};
2  // insert {3, 4, 5} at the back
3  vec.insert(vec.end(), {3, 4, 5});
4  // vec1 now {0, 1, 2, 3, 4, 5}
```

The time complexity of `.insert()` is linear on the number of elements inserted plus the number of the elements after the point of insertion (since each element after this point needs to be shifted over). Similar to `.emplace_back()`, which directly constructs an element at the back of a vector, it is also possible to directly construct an element in the middle of a vector using `.emplace()`.

```
template <typename T, typename... Args>
```

```
iterator std::vector<T>::emplace(const_iterator pos, Args&&... args);
```

Inserts a new element directly before the element at position `pos`. The new element is constructed in place using arguments for its constructor (`args`). An iterator pointing to the emplaced object is returned.

```
1  std::vector<Student> students;
2  students.emplace_back(12345677, "Alice", 3.4);
3  students.emplace_back(12345679, "Cathy", 3.6);
4  students.emplace(students.begin() + 1, 12345678, "Bob", 3.5);
5  // students now stores the students in this order: {Alice, Bob, Cathy}
```

To remove elements from the vector at any position, you should use the `.erase()` member function:

```
template <typename T>
```

```
iterator std::vector<T>::erase(const_iterator position);
```

Erases the element pointed to by the iterator `position` and returns an iterator pointing to the new location of the element that followed the erased element.

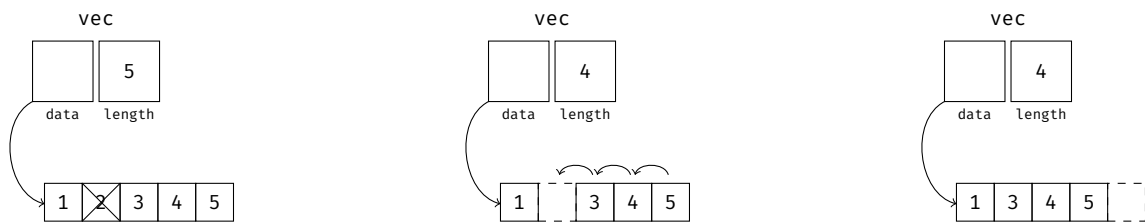
```
template <typename T>
```

```
iterator std::vector<T>::erase(const_iterator first, const_iterator last);
```

Erases all elements in the iterator range `[first, last)` and returns an iterator pointing to the new location of the element that followed the last element erased.

```
1  std::vector<int32_t> vec = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9};
2
3  // erase everything from index 6 to the end
4  vec.erase(vec.begin() + 6, vec.end());
5  // vec now {0, 1, 2, 3, 4, 5}
6
7  // erase element at index 3
8  vec.erase(vec.begin() + 3);
9  // vec now {0, 1, 2, 4, 5}
```

The time complexity of `.erase()` is linear on the number of elements erased plus the number of elements after the erasure point (since elements have to be shifted to ensure there are no gaps in the data).



Both `.insert()` and `.erase()` utilize iterators, which are generalized pointers that can be used to traverse the contents of a container. More detail on iterators will be provided in chapter 11, but the following methods can be used to retrieve iterators for a `std::vector<>` container:

Function	Behavior
<code>.begin()</code>	Returns a random access iterator to the first element in the vector
<code>.end()</code>	Returns a random access iterator to the position one past the last element in the vector
<code>.cbegin()</code>	Returns a <i>constant</i> random access iterator to the first element in the vector
<code>.cend()</code>	Returns a <i>constant</i> random access iterator to the position one past the last element in the vector
<code>.rbegin()</code>	Returns a <i>reverse</i> iterator to the last element in the vector
<code>.rend()</code>	Returns a <i>reverse</i> iterator to the position one before the first element in the vector
<code>.crbegin()</code>	Returns a constant <i>reverse</i> iterator to the last element in the vector
<code>.crend()</code>	Returns a constant <i>reverse</i> iterator to the position one before the first element in the vector

Listed below are some additional operations that are supported by a `std::vector<>`:

Function	Behavior
<code>operator[]</code>	Returns a reference to an element in the vector at a given index
<code>.front()</code>	Returns a reference to the first element in the vector (undefined behavior if empty)
<code>.back()</code>	Returns a reference to the last element in the vector (undefined behavior if empty)
<code>.empty()</code>	Returns whether the vector is empty (true or false)
<code>.clear()</code>	Erases all elements in the vector and leaves it with a size of 0 (but keeps capacity unchanged)

```

1  std::vector<int32_t> vec = {183, 203, 280, 281};
2
3  std::cout << vec.front() << '\n';           // prints 183
4  std::cout << vec.back() << '\n';           // prints 281
5  std::cout << vec[1] << '\n';               // prints 203
6
7  vec.clear();
8  std::cout << vec.empty() << '\n';         // prints 1 (true)

```

7.3 Resize and Reserve

※ 7.3.1 The Costs of Reallocation

A vector stores and manages its data in an array that is dynamically allocated on the heap. The size of a vector represents the number of valid elements that the vector *actually* holds, while the capacity of a vector represents the number of elements the vector's underlying array is *able* to hold. The capacity of the vector's underlying array may be larger than the size of the vector itself, since additional space may be allocated in advance whenever reallocation occurs. To get the size or capacity of the vector, you can use the following member functions:

Function	Behavior
<code>.size()</code>	Returns the size of the vector
<code>.capacity()</code>	Returns the capacity of the vector's underlying data array

The size and capacity of an empty vector both start at 0. Once the first element is pushed back, space is allocated for an underlying array and the capacity becomes 1. From this point on, most `std::vector<>` implementations double their underlying capacity with each reallocation. For example, after another element is pushed back, the data is reallocated to another array and its capacity doubles to 2. Adding the third element would double the capacity to 4, adding the fifth element would double the capacity to 8, adding the ninth element would double the capacity to 16, and so on. This is the behavior that happens if every element were added using `.push_back()`.

If you instead initialize the vector with a non-zero number of elements (either using a range constructor or initializer list), the initial size and capacity are typically set to the number of elements present at initialization. For instance, if you create the following:

```
std::vector<int32_t> vec = {1, 2, 3, 4, 5};
```

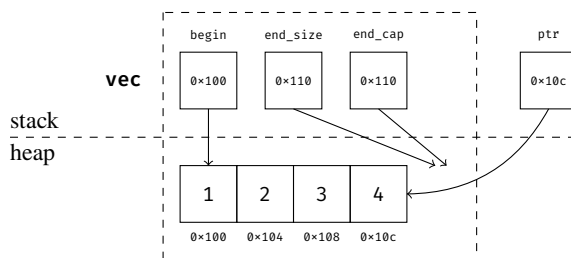
the vector would start off with a size and capacity of 5. Inserting another element into this vector (using `.push_back()`) would increase size to 6, but double capacity to 10. Again, this behavior may depend on the specific standard library implementation, but the distinction between size and capacity remains, as well as the tradeoffs associated with setting capacity differently from the underlying size.

It is important to understand the difference between size and capacity for many reasons. Although a vector's size may be more important when implementing an algorithm or program, failure to consider its capacity could make your code less efficient. Several issues involving capacity and reallocation are summarized below.

1. Reallocation invalidates all existing pointers and iterators to elements in the vector.

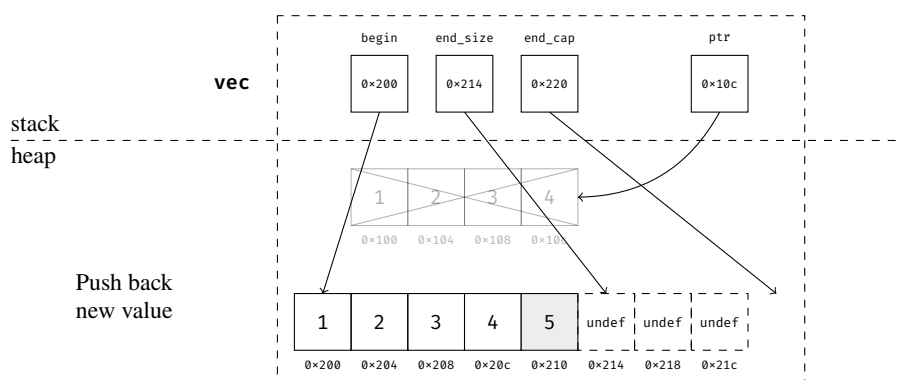
The data in a vector is stored in a fixed-size array on the heap. If the amount of data added exceeds the capacity of the underlying array, the vector has to move the data to a larger array *elsewhere in memory*. Thus, the memory address a pointer or iterator pointed to in the old array would no longer be valid. Consider the following, which initializes a vector of size (and capacity) 4 and creates a pointer to the last element:

```
std::vector<int32_t> vec = {1, 2, 3, 4};
int32_t* ptr = &vec[3];
```



Now, suppose we push back a fifth element to the back of the vector. Since the vector's underlying array only has a capacity of 4, a new larger array must be allocated to hold the fifth element. This triggers a reallocation.

```
vec.push_back(5);
```



Here, the value of `ptr` is the address location `0x10c`, since that was where the element 4 was originally located. However, because of reallocation, the 4 moved in memory — it is no longer at address `0x10c`! Thus, pointers or iterators that were declared before the reallocation are now invalid.²

2. Reallocation takes time.

It takes time to move data to a larger array, since all the elements have to be shifted over from the old array to the new array. If you push back 1,000,000 elements into a vector that doubles capacity during every reallocation, the reallocation process happens 21 times!

3. If you know the intended size of your data, letting the vector set the capacity for you can waste memory.

If you do not tell the vector the size of your data, its default behavior is to double the capacity of the underlying array with every reallocation. This may end up being more than you need! For example, if you wanted to store 1,025 elements in a vector, and you only call `.push_back()` to insert these elements, the vector would double capacity from 1 to 2 to 4 to 8 to 16 to 32 to 64 to 128 to 256 to 512 to 1,024. When you insert the 1,025th element, the vector sees that the underlying array can only hold 1,024 elements, so it doubles capacity to 2,048. The vector doesn't know that element 1,025 was the last one! As a result, even though you only need space for 1,025 elements, you are actually reserving memory for an array of size 2,048 under the hood. All of this remaining memory is unused and wasted.

※ 7.3.2 Reserving a Vector's Capacity

From the three issues above, we can see that reallocation should be avoided whenever possible. If you know the intended size of the vector beforehand, you can prevent these three issues from happening by presetting the capacity of the vector's underlying array upon initialization. For instance, if you know you only need space for 1,025 elements, you could tell the vector to allocate that exact amount of space at the beginning. This can be done using the `.reserve()` member function:

```
template <typename T>
```

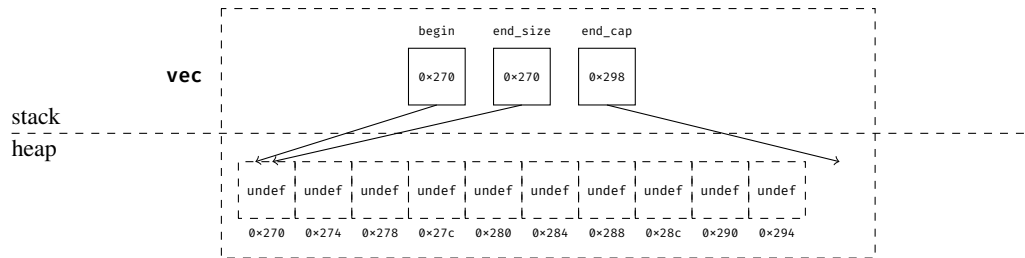
```
void std::vector<T>::reserve(size_t n);
```

Requests that the vector capacity is at least `n`. If `n` is larger than the current vector capacity, this operation causes the vector to reallocate its data to an array with capacity `n`. Otherwise, no reallocation occurs. This function does *not* affect the size of the vector.

²Pointer and iterator invalidation is a pesky bug that is hard to detect. You must be careful when working with pointers or iterators if you modify the contents of the underlying container after creating them. If you need to ensure the stability of references and iterators, you can use containers like the boost library's `boost::container::stable_vector<>` container, which ensures that references and iterators to an element will always remain valid, as long as the element is not erased. Tools like `valgrind`, as covered in chapter 6, can also be used to debug invalidation issues.

It is important to note that the `.reserve()` function does *not* actually change the size of the vector itself! You are *not* creating new elements when you reserve a vector; you are just allocating more space for future elements. For example, the following code reserves a vector of size 10:

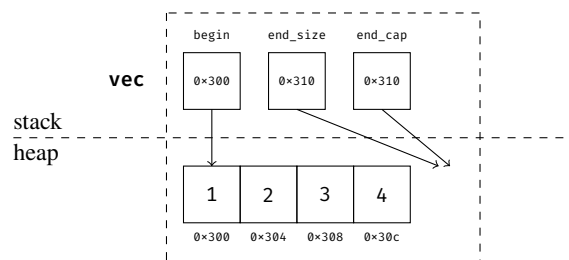
```
std::vector<int32_t> vec;
vec.reserve(10);
```



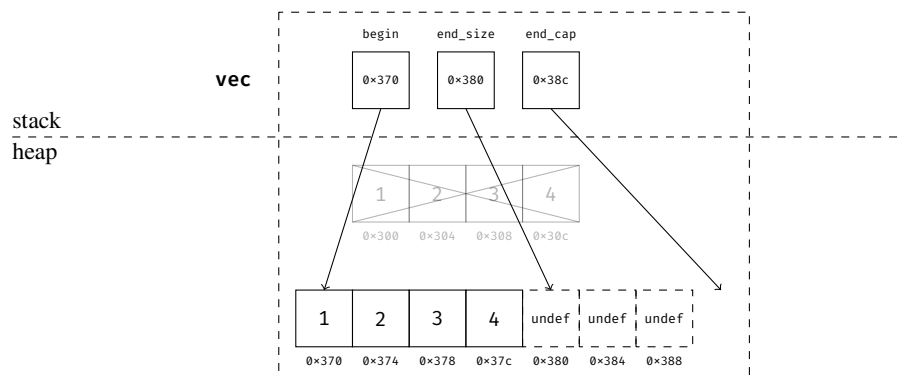
The size of the vector is still 0, since it holds no elements, but the capacity is 10 since the underlying array can hold at most 10 elements. Calling `.push_back()` would add the new element at the first open location (`0x270`).

If you reserve a vector with a capacity that is larger than the current capacity, reallocation occurs. For example, the following code creates a vector that starts at a capacity of 4, but is reserved to a vector of capacity 7.

```
std::vector<int32_t> vec = {1, 2, 3, 4};
```



```
vec.reserve(7);
```



Once again, the size of the vector hasn't changed, but the capacity went up to 7. If you `.push_back()` a new element, it will get added to the first open position (`0x380`).

※ 7.3.3 Resizing a Vector's Size

An alternative would be to use `.resize()`, which sets the *size* of the vector.

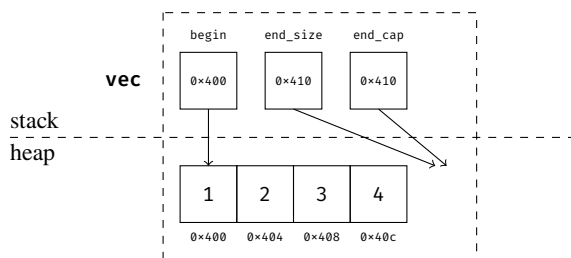
```
template <typename T>
void std::vector<T>::resize(size_t n);
Resizes the vector so that it contains n elements. If n is smaller than the current size, the vector is forced down to a size of n, and any additional elements are destroyed (but capacity remains unchanged). If n is greater than the current size, new elements are value-initialized and added to the end of the array until the vector size becomes n. If n is greater than the current capacity, reallocation occurs, and the capacity also becomes n (see footnote for an exception to this rule).3

template <typename T>
void std::vector<T>::resize(size_t n, T& val);
Same as above, but if n is greater than the current size, copies of val are added to the end of the vector until the vector size becomes n (instead of being value-initialized).
```

³This rule is an oversimplification and is not entirely true in all cases. For the GCC `std::vector<>` implementation at the time of writing, if `n` is larger than the current size, it turns out that capacity only becomes `n` if `n > [2x(previous size)]`. If `n < [2x(previous size)]`, capacity is actually set to `[2x(previous size)]` instead of `n`. That being said, don't go and memorize these rules, since they are implementation-specific and not explicitly stated in the standard.

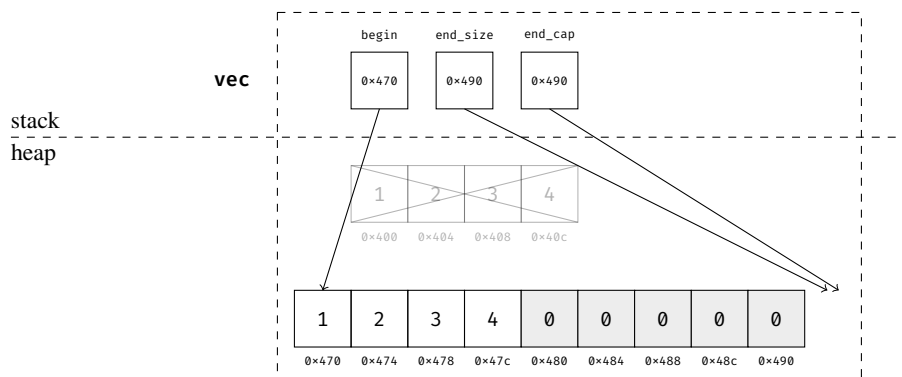
An example using `.resize()` is shown below. A call to `.resize()` actually instantiates data elements in the vector, instead of just allocating more space for the underlying array:

```
std::vector<int32_t> vec = {1, 2, 3, 4};
```



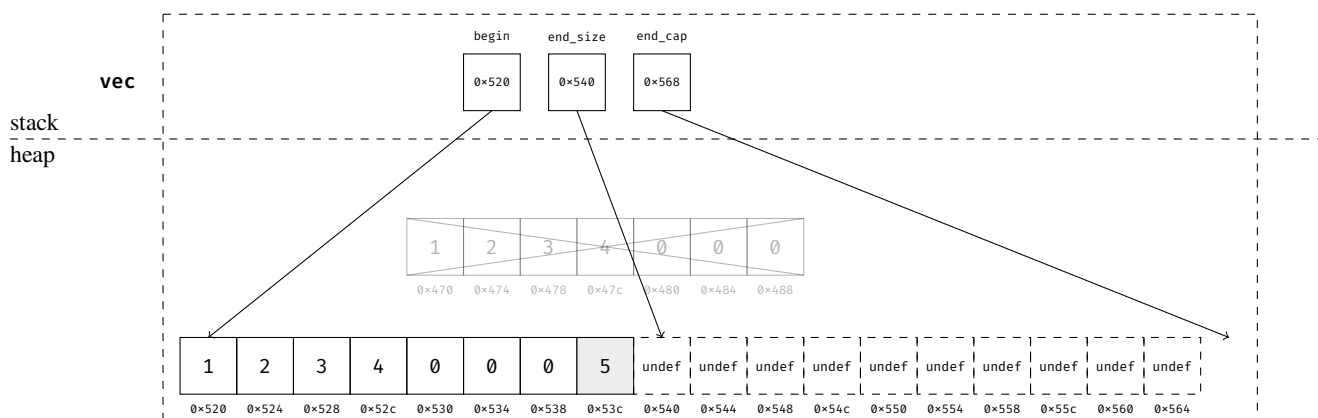
In the example below, the vector is resized to a size of 9 rather than reserved. As a result, the last five elements are value-initialized to 0 (the default behavior for integers), and both size and capacity become 9.

```
vec.resize(9);
```



If we attempt to push back a new element here (e.g., `vec.push_back(5)`), it will not be added to address location `0x480` since an element already exists there (created by `.resize()`). Instead, this element will be added after the ninth element. Since the capacity is only 9, a reallocation must be done to create room for a tenth element. The capacity thus doubles to 18, as shown below.

```
vec.push_back(5);
```



Both `.resize()` and `.reserve()` can be used if you know the size of your vector in advance. Not only can this save memory by ensuring you are only allocating memory you intend to use, it can also speed up your program by reducing the number of reallocations you have to make.

The process of adding new elements differs depending on which approach you use. If you use `.reserve()` to preset the capacity of the vector, you are not creating any elements. As a result, you should use `.push_back()` to add an element to the back of the vector. On the other hand, if you use `.resize()` to preset the size of the vector, you are *creating* elements in the vector. In this case, using `.push_back()` may result in unintended behavior, since it would add an element *after* the values that were created (and may even trigger a reallocation, as shown above). To add values to a vector that has been resized upon initialization, you will have to modify the data values that were created using operator `[]`. An example is shown below:

```
1  std::vector<int32_t> v1;
2  v1.reserve(5);    // reserve capacity to 5
3  v1.push_back(1);  // add elements to back
4  v1.push_back(2);  // using .push_back()
5  v1.push_back(3);
6  v1.push_back(4);
7  v1.push_back(5);

1  std::vector<int32_t> v2;
2  v2.resize(5);     // resize size to 5
3  v2[0] = 1;        // add elements by
4  v2[1] = 2;        // modifying the values
5  v2[2] = 3;        // that were created
6  v2[3] = 4;
7  v2[4] = 5;
```

Remark: For a real life analogy of how `.resize()` and `.reserve()` work, imagine a portable refrigerator filled with water bottles for a sporting event. You have to bring the refrigerator to an event, but you are only allowed to bring a single refrigerator. You also need to ensure that you have enough water bottles on hand to satisfy all demand.

In this scenario, the refrigerator represents the underlying heap-allocated fixed-size array that stores the vector's data. The water bottles represent the data values that are stored. And you, the person carrying the refrigerator around, represents the vector itself — you are responsible for managing the refrigerator and the water bottles inside it. The fact that you can only carry one refrigerator is analogous to the fact that a vector only manages one data array.

When you call `.resize()`, you are changing the number of valid elements in the vector. For example, if you call `vec.resize(50)`, you are setting the number of elements in the vector to 50. This is analogous to adding or removing water bottles from the refrigerator until only 50 water bottles are remaining.

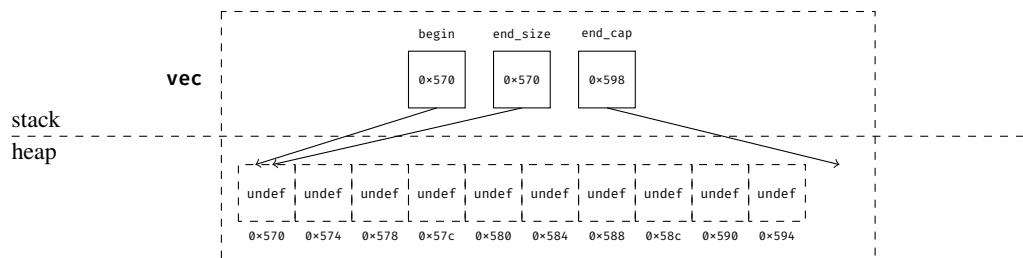
When you call `.reserve()`, you are changing the capacity of the underlying data array, or how many elements the array is able to hold. If you reserve a vector to have a capacity of 50, the vector's underlying array will be allocated to have a capacity of 50. This is analogous to purchasing a refrigerator that can fit a total of 50 water bottles. A call to `.reserve()` does not change a vector's size, much like how buying a bigger fridge does not change the number of water bottles you have.

If you have more water bottles than you have space in your refrigerator, you will need to buy a larger fridge to be able to hold all your water bottles. This is equivalent to allocating a new, larger array in memory. Then, you'll have to move all of your water bottles from the old fridge to the new fridge, and "discard" the old fridge (i.e., not carry it around anymore using our example). This is similar to the process of copying elements over and deallocating the old array.

Example 7.1 Consider the following code. What does line 3 do? Is the value of `b` on line 4 true or false?

```
1  std::vector<int32_t> vec;
2  vec.reserve(10);
3  vec[0] = 5;
4  bool b = vec.empty();
```

Line 3 actually results in undefined behavior. When `.reserve()` is called, it does not actually change the contents of the vector itself, just the capacity of the underlying data array. As a result, the vector is still empty, and the indexing on line 3 ends up accessing uninitialized memory (as there is no element at index 0 yet). Similarly, line 4 assigns `b` to true, since the vector is indeed empty (its size is 0).



Example 7.2 Consider the following code. What does line 4 print?

```
1  std::vector<int32_t> vec;
2  vec.resize(10);
3  vec.push_back(1);
4  std::cout << vec.size() << '\n';
```

When `.resize()` is called, it changes the size of the array. Thus, line 2 sets the size of the array to 10. Line 3 pushes back an element *after* the 10 elements that were created by `.resize()`, so the size becomes 11 (which is printed out on line 4).

※ 7.3.4 Accessing a Vector's Underlying Data Array ※

When you use operator[], you end up retrieving an element from the vector by *reference* (the reasoning for this is explained in section 6.7). However, what if you wanted to return a *pointer* to the underlying array that stores the vector's data (e.g., `0x570` using the addresses in the above diagram)? If you are given a vector `vec`, one idea would be to retrieve the address by calling `&vec`. However, this does *not* give you a pointer to the data stored on the heap. Rather, this gives you a pointer to the vector object that lives on the *stack* (remember that the `std::vector<>` object itself, which is composed of three pointers, lives on the stack — this is not the same as the vector's *data*, which lives on the heap).

To retrieve a pointer to the vector's underlying data array, you will have to call the `.data()` member function instead. Given a vector `vec`, calling `vec.data()` would return a pointer to the the underlying data array of `vec`.

```
template <typename T>
```

```
T* std::vector<T>::data();
```

Returns a pointer to the vector's underlying data array.

Remark: An alternative is to use `&vec[0]`, which also returns a pointer to the underlying data array. However, this may yield undefined behavior if the vector is empty. This is why `.data()` is preferred, as it is always safe to use regardless of whether or not the vector is empty.

7.4 Storing Multidimensional Data in Vectors

Similar to regular C-style arrays, you can use a vector to store multidimensional data. To create a vector that stores two-dimensional data, simply declare a vector that stores other vectors (i.e., a vector of vectors). Here, the outer vector represents one dimension (e.g., rows) while the inner vectors represent another dimension (e.g., columns).

```
std::vector<std::vector<int32_t>> vec_2d;
```

Remark: For these notes, we will refer to a vector of vectors as a "2-D vector" for succinctness (and a vector of vector of vectors as a "3-D vector", etc.). However, it should be emphasized that a multidimensional vector is merely a one-dimensional vector that stores other vectors. For example, even though we will refer to the above vector as a two-dimensional vector, it is actually represented as a one-dimensional vector of `std::vector<int32_t>` behind the scenes, rather than a two-dimensional vector of `int32_t`.

We can extend this definition to multiple dimensions. To declare a three-dimensional vector, simply create a vector object that stores 2-D vectors (where each layer represents a dimension).

```
std::vector<std::vector<std::vector<int32_t>>> vec_3d;
```

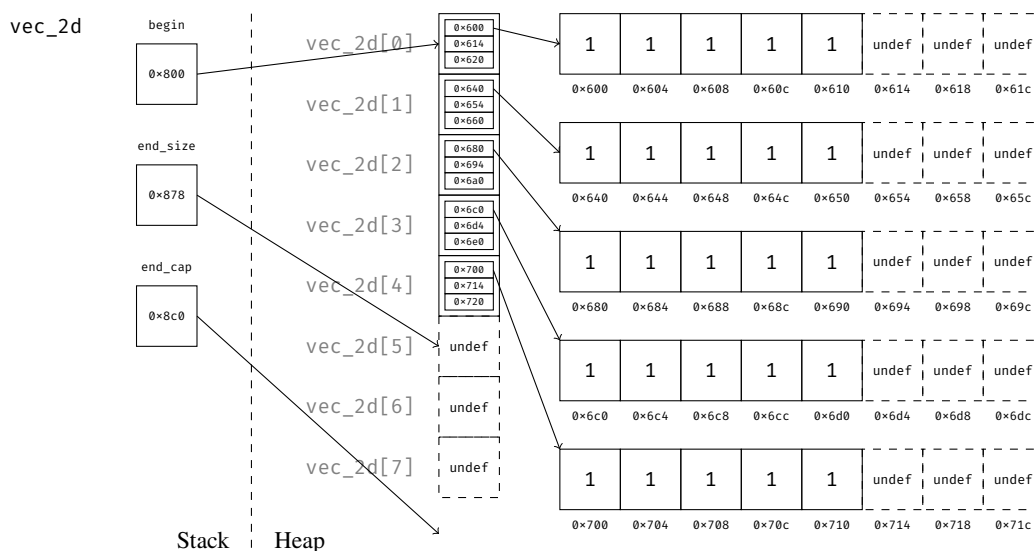
How do we fill the contents of a multidimensional vector? For instance, suppose we wanted to create a 5×5 vector with the following data:

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

A naïve approach would be to use `.push_back()` to insert every element into the 2-D vector. This approach is shown below:

```
1 std::vector<std::vector<int32_t>> vec_2d;
2 while (vec_2d.size() < 5) {
3     std::vector<int32_t> temp;
4     while (temp.size() < 5) {
5         temp.push_back(1);
6     } // while
7     vec_2d.push_back(temp);
8 } // while
```

However, there is an issue with this approach. If we were to look at the vector's underlying memory, we would see something like this (each vector object stores three pointers that each point to a position in its underlying data array):



This ended up happening because we did not specify the size of the vector at the beginning! Thus, the vector automatically doubled with each reallocation, and each dimension ended up having a capacity of 8 instead of 5. To resolve this issue, we must explicitly tell the vector that its size will be 5 using either `.resize()` or `.reserve()`.

```
1 vector<vector<int32_t>> vec_2d;
2 vec_2d.reserve(5);
3 while (vec_2d.size() < 5) {
4     vector<int32_t> temp(5, 1); // resizes vector with 5 elements, each init to 1
5     vec_2d.push_back(temp);
6 } // while
```

However, this is still a better way to do this. It turns out that a multidimensional vector can be declared on a single line using the vector fill constructor. Recall that a vector declared using the following syntax will have a size of `sz`, where each element is initialized to `val`:

```
std::vector<T> vec(size_t sz, T& val);
```

This is exactly what we did on line 4 of the previous code. When we ran the line `vector<int32_t> temp(5, 1)`, we created a vector of size 5, where each element was initialized to 1. The great thing about the fill constructor is that it also sets the *capacity* to the specified size! Because of this, the fill constructor will allow you to declare a vector whose underlying array has exactly the right capacity (e.g., 5 instead of 8).

The `val` in the above fill constructor syntax dictates what our vector is filled with. In the case of a 2-D vector, we want to create a vector of vectors. As a result, we can declare a 2-D vector by setting `val` equal to the constructor of our inner vector, as shown:

```
std::vector<std::vector<T>> vec_2d(m, std::vector<T>(n, data));
```

Here, `m` represents the size of the outer vector, `n` represents the size of the inner vector, and `data` represents the value each element is initialized to. In other words, we are essentially creating an outer vector of size `m`, where each element of this outer vector is initialized to a separate vector of size `n` whose elements are all initialized to `data`. For instance, we can create the previous 5×5 vector by running the following line of code.

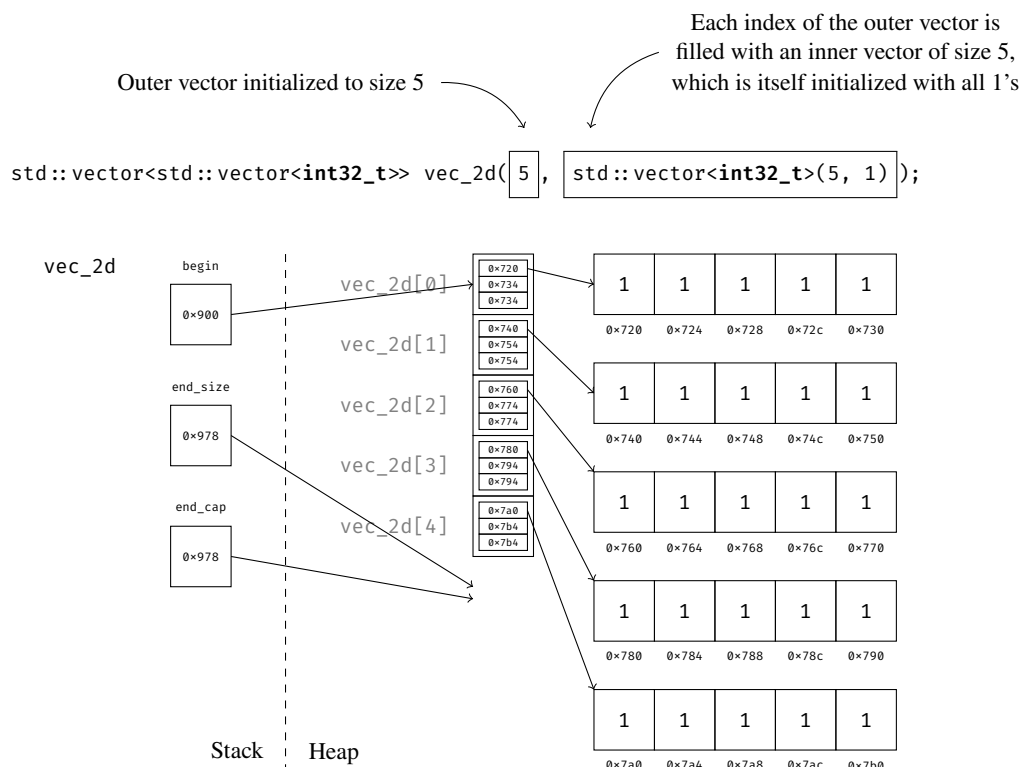
```
std::vector<std::vector<int32_t>> vec_2d(5, std::vector<int32_t>(5, 1));
```

This single line of code does the exact same thing as the previous while loop. This is the preferred way of declaring a 2-D vector, since it is clean, efficient, and ensures that memory is not wasted due to excess capacity. The result of constructing the 2-D vector with the correct size is shown below; unlike the vector in the previous diagram, no memory is wasted.

To initialize vectors of multiple dimensions, simply increase the number of nested vectors. For example, if you want to initialize a 3-D vector with dimensions $2 \times 5 \times 10$ with all elements initialized to 1, you can construct it using the following line of code:

```
std::vector<std::vector<std::vector<int32_t>>> vec_3d(2, std::vector<std::vector<int32_t>>(5, std::vector<int32_t>(10, 1)));
```

To access an element in a multidimensional vector, you can use `operator[]` multiple times (e.g., `vec[2][3]` for the element at row index 2, column index 3 of a 2-D vector). This works because `vec[2]` is also a vector object itself, and thus can be indexed just like `vec`.



7.5 Vector Performance and Memory Overhead

With a 2-D vector, a common convention is to treat the row dimension as the outer vector and the column dimension as the inner vectors (i.e., each index of the outer vector stores a *row* of data). However, the order in which you define your dimensions could potentially affect the performance of data access. Recall that it is faster to sequentially access memory addresses that are closer in memory than ones that are farther apart. Thus, the dimension that you plan on iterating through first should be given the outermost vector in your vector declaration, if possible.



Because the row dimension is typically set as the outer dimension, it is faster to loop through a 2-D vector with rows in an outer loop and columns in an inner loop (under the assumption that rows are the outer dimension). However, different types of problems may require different orders of iteration. For instance, if a problem requires you to visit a 2-D vector column-wise from left to right, it may be preferable to store columns as the outer vector and rows as the inner vector. This allows you to take advantage of faster sequential access for memory in close proximity. If you do this, you will have to make sure to flip the indices when using `operator[]` (i.e., `vec_2d[row][col]` becomes `vec_2d[col][row]`).



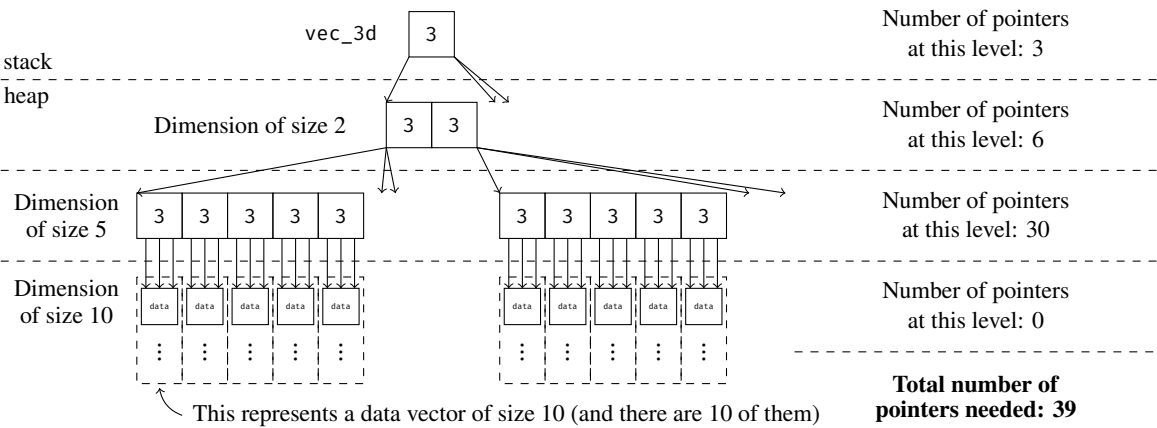
It turns out that the order of dimensions not only affects performance; it may also affect memory usage! This may seem a bit counterintuitive; for instance, if your data has dimensions of $2 \times 5 \times 10$, you would need to store 100 data values in your 3-D vector regardless of whether the outermost dimension is 2 or 5 or 10. Since the number of values you need to store is constant, wouldn't the memory usage be constant as well, regardless of how you declare your vectors?

It turns out that this is not exactly true. Even though the underlying data itself will always take up 400 bytes of memory (assuming the vector is properly resized or reserved), each vector object also stores 3 pointers of bookkeeping data on the stack! The ordering of the dimensions actually changes the total number of pointers needed for our multidimensional vector. Since each pointer takes up 8 bytes of memory on a 64-bit machine, differences in memory usage can quickly accumulate among different dimension orderings.

Let's look at our $2 \times 5 \times 10$ vector again. On the previous page, we declared this vector with 2 as the size of the outermost dimension, 5 as the size of the middle dimension, and 10 as the size of the innermost dimension:

```
std::vector<std::vector<std::vector<int32_t>>> vec_3d(2, std::vector<std::vector<int32_t>>(5, std::vector<int32_t>(10, 1)));
```

How much bookkeeping memory does this use? There is only 1 outer vector, which uses 3 pointers. This outermost vector stores 2 vectors of size 5 (since 2 is the outermost dimension), which use up 2×3 , or 6 pointers. These 2 vectors of size 5 each store a vector of size 10 at each index. Thus, there are 2×5 vectors of size 10, which use up a total of $2 \times 5 \times 3$, or 30 pointers. The total number of pointers needed to declare this $2 \times 5 \times 10$ vector is therefore $3 + 6 + 30$, or 39. The below diagram illustrates the number of pointers needed at each level.



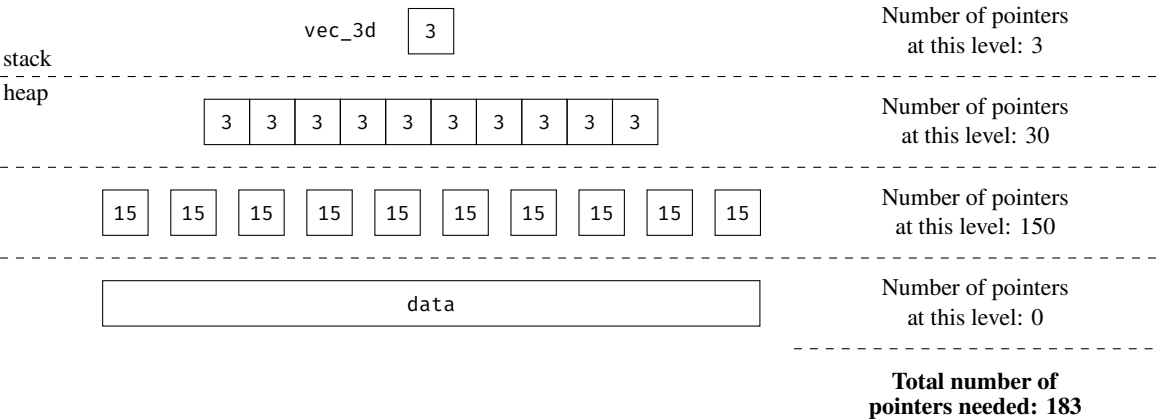
In this diagram, each boxed 3 represents the three pointers of a vector object (i.e., `begin`, `end_size`, and `end_cap`).

The total number of bookkeeping pointers needed for the 3-D vector can be obtained by summing up the number of pointers at each level. The vectors at the lowest layer of the diagram have size 10 (they were not fully drawn out for space reasons), but note that there are no pointers at this level. This is because the vectors at the bottom layer are filled with integers, and not other vector objects.

Let’s see what happens when we flip the dimensions around and declare a $10 \times 5 \times 2$ vector instead.

```
std::vector<std::vector<std::vector<int32_t>>> vec_3d(10, std::vector<std::vector<int32_t>>(5, std::vector<int32_t>(2, 1)));
```

Now, our outermost vector stores 10 vectors of size 5, rather than 2 vectors of size 5 as before. This ends up using 10×3 , or 30 pointers. These 10 vectors of size 5 each store a vector of size 2 at each index, which end up using $10 \times 5 \times 3$, or 150 pointers. Thus, the total number of pointers needed for this 3-D vector is $3 + 30 + 150$, or 183. This is over 4x the number of pointers we needed before we flipped the dimensions!



In the above diagram, each boxed `15` represents a vector of 5 vector objects (each vector uses 3 pointers, so a vector of 5 vectors uses 15 pointers in total). The arrows were omitted due to space constraints, but each of the 10 vectors in the first heap layer points to a vector of size 5, and each vector within the vectors of size 5 points to a data vector of size 2. Like before, the data layer stores integers and not pointers, so there are 0 pointers at this final level.

In general, if you have a three-dimensional vector with dimensions a , b , and c that is declared as follows, where a is the outermost dimension and c is the innermost dimension:

```
std::vector<std::vector<std::vector<T>>> vec_3d(a, std::vector<std::vector<T>>(b, std::vector<T>(c, VAL)));
```

the number of bookkeeping pointers needed is $3 + 3a + 3ab$. Therefore, when defining a multidimensional vector, the dimensions should be ordered from smallest to largest (i.e., $a < b < c$) to minimize additional memory overhead. This is true for dimensions above 3 as well (which can be proven using a similar process).

7.6 Summary of Vector Complexities

In this section, we will summarize the time complexities of several vector operations.

Operation	Average-Case Time	Worst-Case Time
Finding an element	$\Theta(n)$	$\Theta(n)$

The worst-case of finding an element in a vector of size n occurs when the element is the last one in the vector, or if the element cannot be found at all. This requires the search to look at all n elements in the vector. In the average case, the element is somewhere in the middle of the vector, which would require the algorithm to look at approximately $n/2$ elements. This is still $\Theta(n)$ since we can drop the coefficient term of $1/2$.

In the STL, this can be done using `std::find()` in the `<algorithm>` library, which will be covered in chapter 11.

Operation	Average-Case Time	Worst-Case Time
Accessing first element	$\Theta(1)$	$\Theta(1)$

The vector keeps track of a pointer to the first element in its underlying array. Thus, the first element can always be accessed in constant time. In the STL, this can be done using `std::vector::front()`.

Operation	Average-Case Time	Worst-Case Time
Accessing last element	$\Theta(1)$	$\Theta(1)$

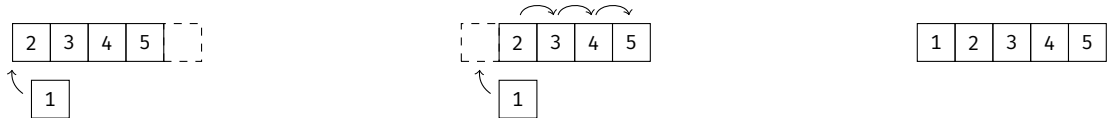
The vector keeps track of a pointer one past the last element in its underlying array. Thus, the last element can always be accessed in constant time. In the STL, this can be done using `std::vector::back()`.

Operation	Average-Case Time	Worst-Case Time
Accessing arbitrary element	$\Theta(1)$	$\Theta(1)$

Since elements in a vector are contiguous in memory, pointer arithmetic can be used to access an element at any index of the vector. Pointer arithmetic takes constant time, so the complexity of accessing an arbitrary element in a vector also takes constant time. In the STL, this can be done using `std::vector::operator[]` or `std::vector::at()` (the latter throws an exception if the given index is out of bounds).

Operation	Average-Case Time	Worst-Case Time
Inserting element at front	$\Theta(n)$	$\Theta(n)$

Since elements in a vector are contiguous in memory, you must shift all elements after the insertion point before you can insert an element. If you attempt to insert an element at the front of a vector of size n , you must shift all n elements after it to the right, which takes $\Theta(n)$ time.



In the STL, this can be done using `std::vector::insert()` or `std::vector::emplace()` with the `begin` iterator.

Operation	Average-Case Time	Worst-Case Time
Inserting element at back	$\Theta(1)$	$\Theta(n)$

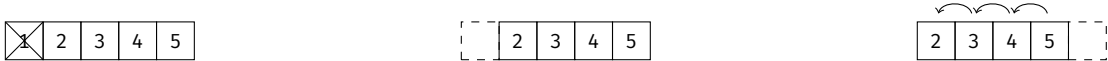
On average, you can insert an element at the back of a vector in constant time, since no elements need to be shifted. However, the worst-case time complexity is $\Theta(n)$ because there is a chance that you will run out of capacity; if this happens, the vector will need to reallocate a new array and copy the existing n elements over. In the STL, this can be done using `std::vector::push_back()` or `std::vector::emplace_back()`.

Operation	Average-Case Time	Worst-Case Time
Inserting element at arbitrary index	$\Theta(n)$	$\Theta(n)$

On average, an insertion will be done in the middle of the vector. Because of this, roughly half of the elements in the vector (i.e., $n/2$ elements) must be shifted to make room for the new element while maintaining the contiguity of elements. This is still considered as $\Theta(n)$ since the coefficient term of $1/2$ is dropped. The worst-case happens if the index is at the front, which requires all n elements to be shifted. In the STL, this can be done using `std::vector::insert()` or `std::vector::emplace()` with an iterator pointing to the position of insertion.

Operation	Average-Case Time	Worst-Case Time
Erasing element at front	$\Theta(n)$	$\Theta(n)$

Much like insertion, erasing an element will require all elements after the erasure point to be shifted to maintain the contiguity of elements. If you attempt to erase an element at the front of a vector of size n , you must shift all n elements after it to the left, which takes $\Theta(n)$ time. This is needed to ensure that the data starts at index 0 and that there are no gaps in the data.



In the STL, this can be done using `std::vector::erase()` with the `begin` iterator.

Operation	Average-Case Time	Worst-Case Time
Erasing element at back	$\Theta(1)$	$\Theta(1)$

If you erase an element at the back of the vector, you do not need to shift anything — the remaining elements are still in their correct positions. The worst-case of an erase at the back is also $\Theta(1)$. This is because, unlike insertion, erasing an element will never trigger a reallocation on its own (so the vector will never allocate a new array and copy elements over). In the STL, this can be done using `std::vector::pop_back()`.

Operation	Average-Case Time	Worst-Case Time
Erasing element at arbitrary index	$\Theta(n)$	$\Theta(n)$

On average, an erasure will be done in the middle of the vector. This requires roughly half of the elements in the vector to be shifted to remove the gap in the data, which is $\Theta(n)$ after dropping the coefficient of $1/2$. The worst case happens if the index is at the front, which requires all n elements to be shifted. In the STL, this is done using `std::vector::erase()` with an iterator (or iterator range) to the value(s) to erase.

Operation	Average-Case Time	Worst-Case Time
Checking if vector is empty	$\Theta(1)$	$\Theta(1)$

This takes constant time by using the vector's pointers to check if the size is 0. In the STL, this is done using `std::vector::empty()`.

Practice Quiz

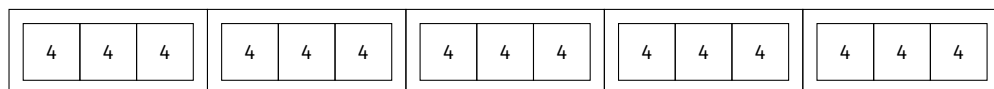
Disclaimer: These practice questions are not official and may not have been vetted for course material. You may use these for practice, but you should prioritize official resources from current staff for a more accurate depiction of what you need to know for assignments and exams.

1. Which of the following statements is/are **TRUE**?
 - I. The actual data elements in a `std::vector<>` are stored on the heap.
 - II. Inserting an element to the back of a `std::vector<>` always takes $\Theta(1)$ time.
 - III. Calling `.reserve()` on a `std::vector<>` never changes the vector's size.
 - A) I only
 - B) II only
 - C) I and II only
 - D) I and III only
 - E) I, II, and III

2. What is the worst-case time complexity of inserting an item at *any* position of a `std::vector<>`, assuming reallocation is needed, and assuming reallocation is not needed?
 - A) $\Theta(1)$ if reallocation is needed, $\Theta(1)$ if reallocation not needed
 - B) $\Theta(1)$ if reallocation is needed, $\Theta(n)$ if reallocation not needed
 - C) $\Theta(n)$ if reallocation is needed, $\Theta(1)$ if reallocation not needed
 - D) $\Theta(n)$ if reallocation is needed, $\Theta(n)$ if reallocation not needed
 - E) None of the above

3. Suppose you are given a program that stores its primary data in a `std::vector<>`. The program executes correctly for small test datasets, but it runs out of memory while reading large datasets. What is a possible solution to overcome this problem?
 - I. Use `.resize()` with the exact number of elements, then use `.push_back()`.
 - II. Use `.resize()` with the exact number of elements, then use `operator[]`.
 - III. Use `.reserve()` with the exact number of elements, then use `.push_back()`.
 - IV. Use `.reserve()` with the exact number of elements, then use `operator[]`.
 - A) I only
 - B) II only
 - C) I and IV only
 - D) II and III only
 - E) I, II, III, and IV

4. Suppose you wanted to instantiate a vector of vectors of integers. The outer vector holds five smaller vectors of integers, each of which individually holds three integer values all initialized to 4.



Which one of the following expressions correctly initializes this vector of vectors?

- A) `std::vector<std::vector<int32_t>> vec(std::vector<int32_t>(5, 4), 3);`
 - B) `std::vector<std::vector<int32_t>> vec(std::vector<int32_t>(3, 4), 5);`
 - C) `std::vector<std::vector<int32_t>> vec(4, std::vector<int32_t>(5, 3));`
 - D) `std::vector<std::vector<int32_t>> vec(5, std::vector<int32_t>(4, 3));`
 - E) `std::vector<std::vector<int32_t>> vec(5, std::vector<int32_t>(3, 4));`
5. If the addition of a new element causes a vector to go over capacity, it has to reallocate. Which of the following is/are **TRUE** regarding the downsides of reallocation?
 - I. Reallocation can be expensive, especially if the elements to be copied are large.
 - II. Reallocation invalidates all existing pointers and iterators to elements in the old underlying array.
 - III. If you know the intended size of your container but do not reserve capacity beforehand, reallocation could result in wasted memory that is allocated but not used.
 - A) I only
 - B) III only
 - C) I and II only
 - D) I and III only
 - E) I, II, and III

6. Suppose you have a container that doubles in capacity and copies all elements to a new location whenever an element is added while the container is at full capacity. You initialize the container's size to 12 and its capacity to 16. Which of the following operations would result in a reallocation? Assume that all additional elements are inserted using `.push_back()`, and that no additional modifications to size or capacity are made.

A) Inserting the 13th element into this container
 B) Inserting the 25th element into this container
 C) Inserting the 33rd element into this container
 D) Inserting the 49th element into this container
 E) More than one of the above

7. How many times does the vector allocate space in this code? Assume that, upon initialization, the vector's capacity is equal to its size.

```
1  int main () {
2      std::vector<int32_t> vec(2);
3      vec.reserve(4);
4      for (int32_t i = 0; i < 7; ++i) {
5          vec.push_back(i);
6      } // for
7  } // main()
```

A) 2
 B) 3
 C) 4
 D) 5
 E) 6

8. Consider the following snippet of code:

```
1  int main () {
2      std::vector<int32_t> vec;
3      vec.reserve(10);
4      for (int32_t i = 0; i < 10; ++i) {
5          vec[i] = 281 + i;
6      } // for i
7      for (int32_t j = 0; j < vec.size(); ++j) {
8          std::cout << vec[j] << " ";
9      } // for j
10     std::cout << '\n';
11 } // main()
```

Running the above code results in undefined behavior. Which line directly causes this behavior?

A) Line 4
 B) Line 5
 C) Line 7
 D) Line 8
 E) None of the above

9. Consider the following snippet of code:

```
1  int main () {
2      std::vector<int32_t> vec;
3      for (int32_t i = 1; i < 10; ++i) {
4          vec.push_back(i);
5      } // for i
6      vec.resize(5);
7      vec.resize(8, 100);
8      vec.resize(9);
9      vec.reserve(10);
10     vec[9] = 10;
11     for (size_t i = 0; i < vec.size(); ++i) {
12         std::cout << vec[i] << ' ';
13     } // for i
14     std::cout << '\n';
15 } // main()
```

Running the above code results in undefined behavior. Which line directly causes this behavior?

A) Line 3
 B) Line 6
 C) Line 7
 D) Line 10
 E) None of the above

10. Consider the following snippet of code:

```
1  int main () {
2      std::vector<int32_t> vec = {1, 2, 3, 4, 5};
3      vec.reserve(6);
4      int32_t* ptr = &vec[1];
5      vec.push_back(6);
6      vec.push_back(7);
7      std::cout << *ptr + vec[3] << std::endl;
8  }
```

What does this code print?

- A) 4
- B) 5
- C) 6
- D) 7
- E) Impossible to determine, since this is undefined behavior

11. Consider the following snippet of code:

```
1  int main () {
2      std::vector<int32_t> v1;
3      std::vector<int32_t> v2;
4      for (int32_t i = 281; i < 381; i += 10) {
5          v1.push_back(i);
6      } // for i
7      v1.resize(6);
8      v2.resize(6);
9      for (int32_t j = 482; j < 494; ++j) {
10         v2.push_back(j);
11     } // for j
12     v1.resize(v2.size(), v2.size());
13     for (int32_t k = 1; k < v1.size(); k += 4) {
14         std::cout << v1[k] + v2[k] << ' ';
15     } // for k
16 }
```

What does this code print?

- A) 281 321 484 488 492
- B) 281 321 502 506 510
- C) 291 331 497 501 505
- D) 291 331 503 507 511
- E) 291 331 504 508 512

12. You want to store a four-dimensional data matrix in the form of a `std::vector<>`. You know that your data has dimensions of $203 \times 281 \times 183 \times 280$. If you want to minimize the amount of memory needed to create the 4-D vector (i.e., a vector of vector of vector of vectors), what should the dimension of the outermost vector be?

- A) 203
- B) 281
- C) 183
- D) 280
- E) It does not matter, since the memory required to create the vector will always be the same

13. Consider the following snippet of code:

```
1  int main () {
2      std::vector<int32_t> v1{2, 5, 1, 3, 4};
3      std::vector<int32_t> v2;
4      for (size_t i = 0; i < v1.size(); ++i) {
5          v2.resize(v1[i]);
6          v2.back() = i;
7      } // for i
8      for (auto num : v2) {
9          std::cout << num << ' ';
10     } // for num
11 }
```

What does this code print?

- A) 2 0 3 4
- B) 2 0 3 4 1
- C) 2 5 1 3
- D) 2 5 1 3 4
- E) Segmentation fault

14. Assume that you have a vector that, when its size is equal to its capacity and a new element is appended using `.push_back()`, all of the elements are copied to a new block of dynamic memory with a new capacity that is twice the previous capacity. When the capacity is 0 and a new element is added, the new container's capacity becomes 1. Suppose you push back n elements into a vector with initial size 0. What is the total number of times that an element is copied from an old block of dynamic memory to a new one?
- $n - 1$
 - $2 \times \sum_{i=0}^{\lfloor \log_2(n-1) \rfloor} i$
 - $\sum_{i=0}^{\lfloor \log_2(n-1) \rfloor} 2^i$
 - $2 \times \sum_{i=0}^{\lfloor \log_2(n-1) \rfloor} i^2$
 - $\sum_{i=0}^{\lfloor \log_2(n-1) \rfloor} 2^i$
15. You are reading in a text file with 1,000,000 words. Because you know in advance how many words you are going to read in, you reserve a vector of strings with a capacity of 1,000,000. However, you make a mistake when reading the file and accidentally insert an additional empty string into the vector after pushing in all 1,000,000 words. Which of the following statements is **FALSE**? Assume that the implementation of this vector is the same as the vector detailed in the previous question.
- Adding in this empty string would cause the vector's memory usage to be roughly twice the memory it would have used had the empty string not been pushed in
 - Adding in this empty string would require your program to copy or move 1,000,000 strings that you would not have needed to copy or move otherwise
 - Adding in this empty string would cause the vector's size to double to 2,000,000 at the end of the program, assuming that no additional elements are pushed in
 - If you had initialized any iterators or pointers to the vector's data prior to pushing in the empty string, all these iterators/pointers would have been invalidated after pushing in the empty string
 - None of the above
16. Which one of the following statements is **FALSE** regarding the `std::vector<>`?
- Inserting new elements into a vector can invalidate pointers and iterators pointing to existing elements in the vector
 - Calling `.resize()` on a vector can invalidate pointers and iterators pointing to existing elements in the vector
 - Calling `.reserve()` on a vector can invalidate pointers and iterators pointing to existing elements in the vector
 - If v is a vector and i is a valid index in v , then $\&v[i] = \&v[0] + i$ will always evaluate to `true`
 - None of the above
17. Suppose you have a vector containing n elements. What is the overall time complexity of appending n more elements, if the vector always reallocates its underlying array with each new element?
- $\Theta(1)$
 - $\Theta(n)$
 - $\Theta(n \log(n))$
 - $\Theta(n^2)$
 - $\Theta(n^3)$
18. Consider the following snippet of code:

```

1  class Thing {
2      int32_t* x;
3      Thing(int32_t x_in) : x{new int32_t{x_in}} {}
4      ~Thing() { delete x; }
5  };
6
7  int main() {
8      int32_t arr[5];
9      std::vector<int32_t> v1;
10     std::vector<Thing> v2;
11     std::vector<int32_t*> v3;
12     std::vector<Thing*> v4;
13     for (int32_t i = 0; i < 5; ++i) {
14         arr[i] = i;
15         v1.push_back(i);
16         v2.push_back(Thing{i});
17         v3.push_back(&arr[i]);
18         v4.push_back(new Thing{i});
19     } // for i
20 } // main()

```

Does this code leak memory, and if so, what is the cause of the leak?

- Yes, the integers pointed to by x in the `Thing` objects of $v2$ are leaked
- Yes, the integers pointed to by the elements of $v3$ are leaked
- Yes, the `Thing` objects pointed to by the elements of $v4$ are leaked
- More than one of the above
- No memory is leaked after the vector destructors are run

19. You are given a vector of integers `nums`. Implement a function that returns the smallest index `i` of `nums` such that the last digit of `nums[i]` matches the last digit of its index `i` (i.e., `nums[i] mod 10 == i mod 10`). If no such index exists, return `-1`. For example, given the following vector:

32	46	10	53	24	67	98
0	1	2	3	4	5	6

you would return 3, since that is the first index whose last digit matches the last digit of its value (e.g., 53 at index 3).

```
int32_t smallest_equal_index(const std::vector<int32_t>& nums);
```

You should implement your solution in worst-case $\Theta(n)$ time and $\Theta(1)$ auxiliary space, where n is the size of the vector. Note that you can use operator `%` to calculate a modulus (e.g., `53 % 10 = 3`).

20. You are given an $n \times n$ square matrix of integers in the form of a vector of vectors. Implement a function that returns the sum of the values on the matrix diagonals (each value is only counted once, even if it is on two diagonals). For example, given the following matrix, you would return 25, since that is the sum of the diagonals ($1 + 3 + 5 + 7 + 9 = 25$).

1	2	3
4	5	6
7	8	9

1	2	3
4	5	6
7	8	9

```
int32_t matrix_diagonal_sum(const std::vector<std::vector<int32_t>>& matrix);
```

You should implement your solution in worst-case $\Theta(n)$ time and $\Theta(1)$ auxiliary space, where n is the size of a matrix dimension.

21. In linear algebra, a Toeplitz matrix is a matrix where each descending diagonal from top-left to bottom-right contains the same value. For example, the following matrix is a Toeplitz matrix because each top-left to bottom-right diagonal consists of the same elements.

1	2	3	4
0	1	2	3
2	0	1	2

1	2	3	4
0	1	2	3
2	0	1	2

Given a $m \times n$ matrix in the form of a vector of vectors, implement a function that returns whether the matrix is a Toeplitz matrix or not. If a matrix is Toeplitz, return `true`; otherwise, return `false`.

```
bool is_toeplitz(const std::vector<std::vector<int32_t>>& matrix);
```

You should implement your solution in worst-case $\Theta(mn)$ time and $\Theta(1)$ auxiliary space, where m and n are the dimensions of the matrix.

» « Quiz Solutions » «

1. **The correct answer is (D).** Only statements I and III are true. Statement I is true because the data in a vector are stored in an underlying array that is allocated on the heap. Statement III is true because `.reserve()` only affects the vector's capacity, and not its size. Statement II is false because you may have to reallocate if the new element causes the vector to go over capacity, which would take linear time.
2. **The correct answer is (D).** The worst-case time complexity of insertion assuming reallocation is $\Theta(n)$, since you would have to move all the existing data to a new underlying array. Without reallocation, however, the worst-case time complexity could still be $\Theta(n)$ if the insertion takes place at the front of the vector, which would require all subsequent elements to be shifted over by one position.
3. **The correct answer is (D).** When you call `.resize()` on a vector, you actually change the number of elements it contains. On the other hand, if you call `.reserve()`, you only reserve space for more elements, but you do not create them. Hence, you cannot use `.push_back()` on a vector that has been resized, since you would be adding elements on top of the elements you already created using `.resize()` (i.e., if you want 20 elements in your vector and nothing more, resizing the vector would initialize 20 elements in the vector, and using `.push_back()` would add a 21st element instead of modifying the 1st element). Meanwhile, after reserving space for a vector to hold more elements, you still must add the elements themselves. For instance, if you reserve a vector to hold 20 elements, it now has the capacity to hold 20 elements, but it does not actually have any elements yet (similar to how an empty 5-gallon bucket *can* hold 5 gallons of water, but it does not actually hold any water). Thus, you cannot use `operator[]` on a vector that only has space reserved, as the elements you want to create do not exist yet. The only ways to ensure that a vector has the number of elements you want it to have are to either resize the vector and modify each element using `operator[]` or reserve space for the vector and add the elements using `.push_back()`.
4. **The correct answer is (E).** When initializing a vector using a fill constructor, the first argument is the intended size of the vector, and the second argument is the value you want initialized for each of the elements in the vector. Only choice (E) works here: you are initializing a vector of size 5 (first argument) where each element is initialized to a vector of integers (second argument). This inner vector is given a size of 3 (first argument), where each element is initialized to 4 (second argument).
5. **The correct answer is (E).** All of the statements are true. Statement I is true because elements must be copied from one array to another during reallocation, which can be an expensive process. Statement II is true because all pointer or iterator references to the old array would be invalidated when the elements are copied to a new, larger array. Statement III is true because you may end up with a larger capacity than you need, which would waste memory (e.g., you only need to store 65 elements, but your vector ends up with a capacity of 128 since capacity is doubled with each reallocation, leaving 63 slots unused).
6. **The correct answer is (C).** The vector reallocates with double the capacity when you try to insert an element that exceeds its capacity. Currently, the capacity is 16, so a reallocation is triggered when you insert the 17th element, which doubles the capacity to 32. The next reallocation happens when you try to insert the 33rd element, which doubles the capacity to 64. The next reallocation happens when you try to insert the 65th element, which doubles the capacity to 128 (and so on).
7. **The correct answer is (C).** The vector allocates space four times. The first allocation occurs on line 2, where the vector's underlying array is initially set to a size of 2 (with the values initialized to zero). The second allocation occurs on line 3, since the requested capacity of 4 is larger than the existing capacity of 2. This is what the vector looks like after the second allocation:

0	0	undef	undef
---	---	-------	-------

Next, the loop on line 4 inserts 7 elements to the back of the vector. Pushing back the 3rd of these elements forces a reallocation since there would be 5 elements after the insertion, but only an existing capacity of 4. This reallocation doubles the vector's capacity to 8. Following this, the insertion of the 7th element in this loop increases the size of the vector to 9, which forces another reallocation of the vector from a capacity of 8 to a capacity of 16.

8. **The correct answer is (B).** The `.reserve()` call on line 2 reserves the vector's capacity to 10, but it does not actually create any elements. Thus, trying to index into the values on line 5 causes undefined behavior, since you cannot index a vector with a position that is larger than its size (which is still zero).
9. **The correct answer is (D).** On line 10, the vector only has a size of 9 (from the `resize` on line 8). Since vectors are zero-indexed, there is no value at index 9, so the use of `operator[]` on line 10 results in undefined behavior.
10. **The correct answer is (E).** It is impossible to determine what line 7 prints, since `ptr` is invalidated when the vector's underlying array is forced to reallocate on line 6 (since a 7th element is added when the capacity is 6).
11. **The correct answer is (D).** The loop on line 4 pushes back all numbers from 281 to 381 (exclusive), incrementing by 10 each time. After the loop runs to completion, the contents of `v1` are:

v1	281	291	301	311	321	331	341	351	361	371
----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

On lines 7 and 8, both `v1` and `v2` are resized to a size of 6. This removes all but the first six elements in `v1` and initializes six elements in `v2` with a default value of zero.

v1	281	291	301	311	321	331
v2	0	0	0	0	0	0

The loop on line 9 pushes back all values from 482 to 494 (exclusive) to the back of v2.

v1	281	291	301	311	321	331													
v2	0	0	0	0	0	0	482	483	484	485	486	487	488	489	490	491	492	493	

Line 12 resizes v1 to the same size as v2, with values initialized to the size of v2 (which is 18).

v1	281	291	301	311	321	331	18	18	18	18	18	18	18	18	18	18	18	18	18
v2	0	0	0	0	0	0	482	483	484	485	486	487	488	489	490	491	492	493	

The loop on line 13 then prints out the sum of the two vectors at indices 1, 5, 9, 13, and 17.

v1	281	291	301	311	321	331	18	18	18	18	18	18	18	18	18	18	18	18	18
v2	0	0	0	0	0	0	482	483	484	485	486	487	488	489	490	491	492	493	
		291				331				503				507					511

12. **The correct answer is (C).** If you have a four-dimensional vector that is declared in the order a , b , c , and d , the total number of bookkeeping pointers needed can be estimated using $3 + 3a + 3ab + 3abc$. Thus, to reduce the total number of pointers you need to keep track of, you should declare the outermost vector (in this case, the one with dimension a) using the size corresponding to the smallest dimension, which in this case is 183.
13. **The correct answer is (A).** For each index i in the loop on line 4, v2 is resized to a value of $v1[i]$ and has its last element assigned to i . On the first iteration, v2 is resized to a size of $v1[0]$ or 2, and the last element is assigned to a value of 0.

v2	0	0
----	---	---

On the next iteration, v2 is resized to a size of $v1[1]$ or 5, and the last element is assigned a value of 1.

v2	0	0	0	0	1
----	---	---	---	---	---

On the next iteration, v2 is resized to a size of $v1[2]$ or 1, and the last element is assigned with a value of 2.

v2	2
----	---

On the next iteration, v2 is resized to a size of $v1[3]$ or 3, and the last element is assigned with a value of 3.

v2	2	0	3
----	---	---	---

On the next and final iteration, v2 is resized to a size of $v1[4]$ or 4, and the last element is assigned with a value of 4.

v2	2	0	3	4
----	---	---	---	---

The contents of this vector are then printed out.

14. **The correct answer is (E).** When the 2nd element is inserted, the vector reallocates and 1 element is copied over. When the 3rd element is inserted, the vector reallocates and 2 elements are copied over. When the 5th element is inserted, the vector reallocates and 4 elements are copied over. When the 9th element is inserted, the vector reallocates and 8 elements are copied over. Since $\log_2(n-1)$ reallocations are done if n elements are inserted without explicitly reserving capacity (since that is the number of times the capacity can double before you can support n elements), the total number of times an element is copied can be expressed using the summation $1 + 2 + 4 + \dots + 2^{\lfloor \log_2(n-1) \rfloor}$, which matches option (E).
15. **The correct answer is (C).** Options (A), (B), and (D) are all true, since these are consequences of vector reallocation, which would happen if you pushed in a 1,000,001st string into a vector whose capacity is 1,000,000. Option (C) is incorrect because the size of the vector is not doubled with reallocation (as with capacity), so the size of the vector would end up being 1,000,001 instead of 2,000,000.
16. **The correct answer is (E).** All of the statements are true. Options (A), (B), and (C) could all result in a reallocation if the new requested size or capacity exceeds the existing capacity, which would result in pointer and iterator invalidation. Option (D) is true because elements in a vector are stored contiguously in memory, allowing us to use pointer arithmetic to identify the position of elements in the vector.
17. **The correct answer is (D).** If the vector always reallocates with each new element added, the first new element added would force the existing n elements to be copied; the second new element added would force the existing $n+1$ elements to be copied; and so on. From this, you can see that each new insertion would take $\Theta(n)$ time for the old elements that need to be copied during reallocation. Since we are appending n new elements, this process is done a total of n times, for an overall time complexity of $n \times \Theta(n) = \Theta(n^2)$.

18. **The correct answer is (C).** Even though vectors store their values on the heap, they are responsible for cleaning up all memory that they own. Thus, the values in `v1` and `v2` are automatically cleaned up their respective vector is destructed. The pointers in `v3` refer to the values in `arr`, which are automatically cleaned up as well when `arr` goes out of scope. This is not the case for the values in `v4`, however, since they are explicitly allocated on the heap by the programmer. Since the programmer created these `Thing` entities using `new`, they are also responsible for cleaning them up using `delete`; this is not done, so the `Thing` objects in `v4` are leaked.
19. This problem can be solved using an iteration of the vector, checking each value to see if its last digit matches its index. Once we encounter such a value, we can return its index immediately. If we finish iterating the entire vector without encountering a value whose last digit matches its index, we would return `-1`. Since we only need to iterate over the vector of n elements once, and we perform a constant operation at each element (checking its last digit with its index), this solution takes $\Theta(n)$ time.

```

1  int32_t smallest_equal_index(const std::vector<int32_t>& nums) {
2      for (size_t i = 0; i < nums.size(); ++i) {
3          if (i % 10 == nums[i] % 10) {
4              return i;
5          } // if
6      } // for i
7      return -1;
8  } // smallest_equal_index()

```

20. One strategy for solving this problem is to use a nested `for` loop, one that iterates over the rows and one that iterates over the columns, and add the values where the row and column index identify a diagonal. However, that would take $\Theta(n^2)$ time, which exceeds the time complexity specified by the problem. Instead, we can implement a linear-time solution by only iterating over the rows and using arithmetic to determine which two values in that row are on a diagonal. For each row index i , the value on a diagonal must be located at column i and $n - i - 1$. One thing to consider is the possibility of duplicating the center value if the dimension of the matrix is odd, so we will also need to subtract the center value in this scenario. One potential implementation is shown below:

```

1  int32_t matrix_diagonal_sum(const std::vector<std::vector<int32_t>>& matrix) {
2      int32_t sum = 0;
3      for (size_t i = 0; i < matrix.size(); ++i) {
4          // value on top-left to bottom-right diagonal
5          sum += matrix[i][i];
6          // value on top-right to bottom-left diagonal
7          sum += matrix[i][n - i - 1];
8      } // for i
9
10     // if dimension is odd, subtract center value so it is not double-counted
11     if (matrix.size() % 2 == 1) {
12         return sum - matrix[n / 2][n / 2];
13     } // if
14
15     return sum;
16 } // matrix_diagonal_sum()

```

21. One strategy for solving this problem is to use a nested `for` loop, one that iterates over the rows and one that iterates over the columns, and check if `matrix[i][j]` equals `matrix[i - 1][j - 1]` for all i and j from 1 to $n - 1$. One implementation is shown below:

```

1  bool is_toeplitz(const std::vector<std::vector<int32_t>>& matrix) {
2      for (size_t i = 1; i < matrix.size(); ++i) {
3          for (size_t j = 1; j < matrix[i].size(); ++j) {
4              if (matrix[i][j] != matrix[i - 1][j - 1]) {
5                  return false;
6              } // if
7          } // for j
8      } // for i
9      return true;
10 } // is_toeplitz()

```